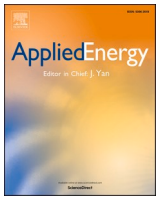
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COA-CNN-LSTM: Coati optimization algorithm-based hybrid deep learning model for PV/wind power forecasting in smart grid applications

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HIGHLIGHTS

- Propose COA-CNN-LSTM model: Coati Optimization Algorithm-based hybrid deep learning CNN-LSTM.
- Solar and wind powers are dependent upon environmental factors, making the forecasting problem very difficult.
- COA-CNN-LSTM outperforms other techniques in terms of the Granger causality test and Natch Shitcliff analysis.
- Experiments show precise and definitive wind power-making predictions for the management of RES.

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ABSTRACT

Power prediction is now a crucial part of contemporary energy management systems, which is important for the organization and administration of renewable resources. Solar and wind powers are highly dependent upon environmental factors, such as wind speed, temperature, and humidity, making the forecasting problem extremely difficult. The suggested composite model incorporates Long Short-Term Memory (LSTM) and Swarm Intelligence (SI) optimization algorithms to produce a framework that can precisely estimate offshore wind output in the short term, addressing the discrepancies and limits of conventional estimation methods. The Coati optimization algorithm enhances the hyper parameters CNN-LSTM. Optimum hyper parameters improvise learning rate and performance. The day-ahead and hour-ahead short-term predictions RMSE can be decreased by 0.5% and 5.8%, respectively. Compared to GWO-CNN-LSTM, LSTM, CNN, and PSO-CNN-LSTM models, the proposed technique achieves an nMAE of 4.6%, RE 27% and nRMSE of 6.2%. COA-CNN-LSTM outperforms existing techniques in terms of the Granger causality test and Nash-Sutcliffe metric analysis for time series forecasting performance, scores are 0.0992 and 0.98, respectively. Experimental results show precise and definitive wind power-making predictions for the management of renewable energy conversion networks. The presented model contributes positively to the body of knowledge and development of clean energy.

1. Introduction

Fossil fuels have long been used to fulfill the energy requirements of the world's increasing population and boost economic growth. However, in recent years, the negative consequences of using fossil fuels have grown obviously, especially the damaging outcomes in terms of air pollution [1]. The airborne pollutants from burning fossil fuels might impair a country's ability to expand responsibly and lower the earth's load capacity. Furthermore, using fossil fuels has an adverse impact on the environment. Because of their market and price uncertainty, supply risk, and uneven distribution, they engage in massive energy instabilities. Thus, the demand for clean energy has grown notably during the last few decades. Renewable energy has gained appreciable recognition as the energy evolution, seen growth because of its environment-friendly features, such as the low amount of carbon, better air quality, and other advantageous attributes [2]. Solar energy along with wind power has surfaced as two of the most efficient energy tools as

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Nomenclature

Acronyms

ANN	Artificial Neural Networks
CNN	Convolutional Neural networks
COA	Coati Optimization Algorithm
DNN	Deep Neural Networks
GA	Genetic Algorithm
GCT	Granger Causality Test
GLSTM	Genetic Algorithm with LSTM
GRU	Gated Recurrent Unit
GWO	Grey Wolf Optimization
ILSTM	Improved Long Short-Term Memory
LM	Local Maxima
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MARE	Mean Absolute Relative Error
NN	Neural Network
OLS	Ordinary Least Squares
PSO	Particle Swarm Optimization
PV	Photo-Voltaic
RE	Relative Error
RES	Renewable Energy Sources
RNN	Recurrent Neural Networks
RMSE	Root Mean Square Error
SI	Swarm Intelligence
SR	Success Rate
STC	Standard Testing Condition
STCNs	Spatiotemporal Convolutional Networks
SVM	Support Vector Machine
WECS	Wind Energy Conversion System

Variables

G_{STC}	Irradiance at STC condition ($1000W/m^2$)
G	Irradiance (W/m^2)
T_{STC}	The temperature at STC condition (25°C)
T	Temperature (°C)
m	Number of decision variables
A_x	Location on search member
lb_y, ub_y	Lower and Upper boundaries of search space
F_x	Objective function for decision making
$F \bullet F$	Fitness Function
T_i	True value
P_i	Predicted value
n	Sample size
$\bar{T}_a$	Average of true output
C_n	Cell state
f_n	Forget gate
O_n	Output gate
σ	Sigmoid activation function

substitutes to cater to energy demands because of several advantages associated with these clean forms of energy, such as being sustainable, clean, and environment friendly [3].

Many researchers have been working to construct highly-precise systems for predicting resources, such as wind and solar energies maintain and protect the longevity of energy systems [4]. Accordingly, the energy utilization of renewable resources is a challenge in terms of social and economic advancements due to the nonlinear behavior of energy resources [5]. Renewable energy sources became increasingly important in the smart grid because they provide clean and abundant energy [6]. However, their variable and unpredictable nature presents significant challenges to the stability and reliability of the grid. For balancing the supply and demand of electricity, RES production must be linked with energy storage, demand response, and other distributed energy resources. Therefore, the system requires advanced control and optimization algorithms that manage various smart grid components to maintain a stable and secure energy supply [7]. In addition, RES forecasting is essential in the smart grid to predict their output accurately. Practical applications of power forecasting in the smart grid can be of four types. Firstly, energy management allows the grid operator to plan and adjust the energy supply to match the demand reliably. By accurately predicting the RES output, the grid operator can adjust the energy supply in real time, reduce the risk of blackouts and improve the reliability of the grid. Second, grid planning in the long term for future capacity requirements makes informed investment decisions. Third, is the energy market operations to help set prices and enable the trading of renewable energy credits. Fourth and most important, is risk management by identifying potential supply disruptions and taking appropriate measures to minimize their impact. By predicting the RES output, utilities can plan for contingencies, such as extreme weather events to reduce the risk of blackouts and ensure that the energy supply remains stable [8].

The main objective of spatiotemporal (ST) studies regarding wind energy is to investigate the spatiotemporal patterns of power generation and its variability in different locations and periods [9]. Wind power generation depends on various factors, such as air temperature, the direction of the wind, atmosphere, wind speed, and dampness. Several elements can vary spatially and temporally, leading to spatiotemporal variability in wind power generation. By analyzing spatiotemporal patterns of wind power, researchers aim to understand the underlying physical mechanisms that govern wind power generation and its variability [10]. Thus, this will be helpful with the layout and optimization of wind power systems and create more precise and trustworthy wind and solar power forecasting models. Large amounts of wind power data are gathered from many places over various periods and often analyzed using advanced statistical and machine learning approaches in spatiotemporal studies [11]. Those techniques can help identifying ST patterns and trends in wind power generation and quantify the effect of different strands on wind power variability [12]. Spatial and temporal analysis using the multi-output support vector machine (MSVM) is carried out. To calculate the wind power output from many farms, the ST-GWO-MSVM model, which uses GWO as a stochastic optimizer, is used [13]. In [14], the asymmetric spatial correlation at various temporal scales is described using the Spatiotemporal Convolutional Networks (STCNs), which use the convolutional directed graph structure to define the adaptive characteristics.

Power generation depends on different attributes of wind energy. Wind velocity is a significant variable for the anticipation of the overall amount of generated power. For solar energy, irradiance plays an important role to estimate the total amount of power generation [15]. Consequently, explicit prognosticative models can aid companies to get the maximum amount of energy from these resources effectively. Thus, decrease their expenses, usually utilize for the nonlinear behavior of energy resources. Many scholars have proposed numerous model types that forecast power generation using various characteristics related to energy resources [16,17]. Forecasting techniques can help get predictions using the important features mainly responsible for the nonlinear behavior of renewable resources. Applying different AI-based techniques, we can predict the underlying relationship between independent and dependent features, which, in turn, affect power generation over time [18]. Hence, attain clean and productive energy from these resources [19–21].

The research objective is to propose a highly efficient power forecasting model that is effective for both wind and solar power in the short

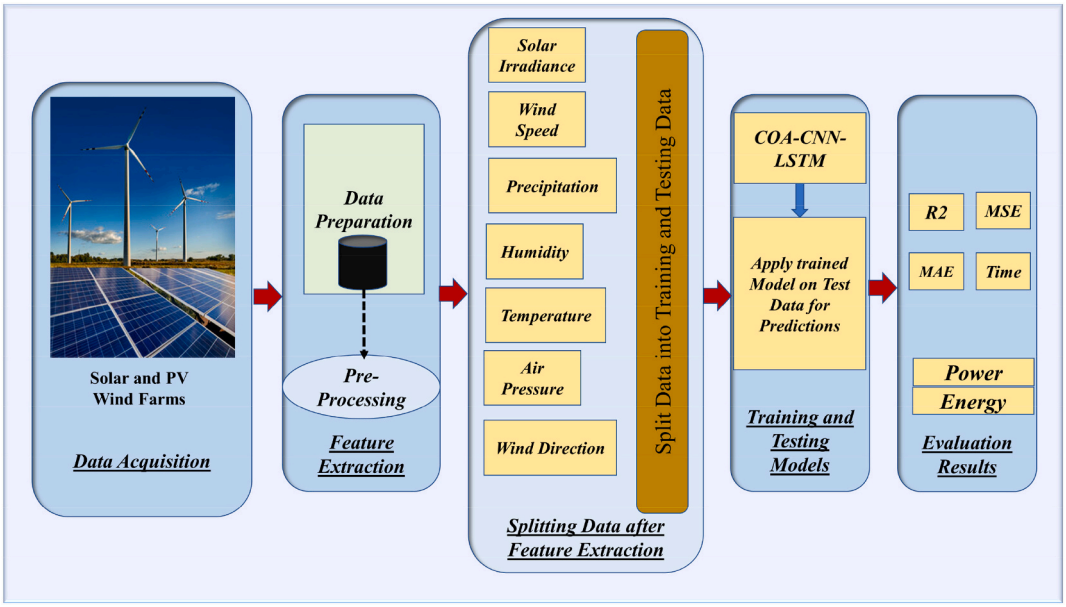


Fig. 1. Graphical presentation of procedure COA-CNN-LSTM based PV/Wind Power Forecasting.

and medium terms. The suggested solar/wind power forecasting system has two main components: LSTM and SI optimization algorithms. Recurrent neural networks (RNNs), such as the LSTM model, are well suited for time-series forecasting problems because they are capable of efficiently capturing long-term relationships and patterns in sequential data. To train the LSTM model, the raw data set composed of historical wind or solar power along with meteorological information is used. SI optimization strategies are used to tune the LSTM model's hyper parameters. These algorithms are a class of stochastic optimization techniques that mimic the collective behavior of social animals like ants and bees to figure out the optimal solution in a lookup field. The potency of the suggested system is assessed using a unique dataset that includes wind energy and weather information from several wind farms. The data values are preprocessed and divided into validation, training, and testing sets of values. Then, the suggested system is used to get insights from the training set. Next, the information is validated through the validation set to tune the optimal hyperparameters using the proposed SI algorithms. Several statistical performance metrics are used to measure the performance of the presented model with that of three modern models, including CNN-LSTM, LSTM, and combinations of CNN, PSO, and GWO.

1.1. Related work

The researchers used historic spatial and climatic data to figure out wind speed, irradiance, and other important features to predict power generation while using different RESs. However, they are not very effective at estimation of current wind speed. Certain policies are utilized in one method to determine if the dependent and independent variables in the short-term wind and solar power predictions are related [22]. These procedures are included in statistical methods. As these models can comprehend the crucial physical properties of the many accessible features, they surpass conventional algorithms in predicting wind speed or solar irradiation. But they cannot spot the nonlinear behavior of different renewable energy resources effectively. A different strategy is a geographical correlation, which is more effective at predicting dependability [23]. In short-term predictions, support vector machine (SVM), fuzzy logic, and more sophisticated approaches of Neural Networks (NNs), such as CNNs, can generate better outcomes. On the other hand, for predicting short-term wind speeds or solar irradiance, Deep Neural Networks (DNNs) outrun other typical machine learning techniques by a wide margin.

Deep learning techniques and other meta-heuristic optimizers have manifested an extraordinary predictive effect on statistical prediction because the improved LSTM is integrated owing to its dynamic character that can be utilized to inspect keynotes in data strings with latencies and longer time spans. In [24], the authors suggested the LSTM-improved forget-gate network algorithm, also considered as LSTM-EFG, to forecast the power generation from clean energy resources. The Genetic Algorithm (GA), a categorization model-based method, is suggested to provide a unique forecasting model [25]. For challenges associated with predicting software vulnerabilities, Genetic Algorithm uses a deep SYMbiotic-based developmental procedure that phenotypes prominent traits. The method enhanced the capacity to identify susceptibility trends in programs with flaws [26].

Using these decomposition techniques enables a more thorough depiction of the fundamental properties of the wind energy signal. The improved LSTM (ILSTM) model is better able to grasp and describe the complex dynamics and patterns involved in wind energy generation because it captures the protracted (long-lasting), oscillating (repetitive), and stochastic (random) components independently [27]. In addition, steps are taken to stop the model's propensity for overfitting the training set of data to solve the overfitting issue. A more reliable and accurate prediction model was created by reducing overfitting. Thus, guarantee that, even in the face of actual atmospheric fluctuations, the model can generalize effectively to previously unobserved data and produce accurate predictions. In [28], the suggested strategy intends to improve the efficiency of hyperparameter optimization by combining these two optimization strategies. The exploration-exploitation balance of the sine-cosine optimization approach facilitates effective investigation of the hyperparameter search space. In the meantime, the search procedure is modified and refined by the self-adaptive differential evolution approach based on the caliber of solutions discovered, improving convergence and optimization performance.

In [29], a genetic architecture, known as GLSTM, consisting of an LSTM model with a genetic algorithm (GA) is proposed. GLSTM can forecast short-term predictions for solar and wind power production. The architecture of dynamic algorithms like LSTM is proposed here because of its strength to perform well on continuous inputs, which depend on timestamps. On the contrary, optimization technique, such as GA is utilized to exaggerate the tally of windows and units' size, which is used in LSTM. After that, the GLSTM findings are contrasted with the

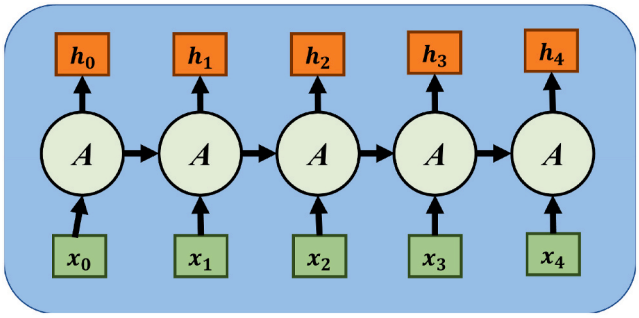


Fig. 2. Structure of RNN model.

real power, established methods used in conventional performance measures, and linear regression. The GLSTM findings were higher by 6 to 30%. The GLSTM and LSTM differ significantly in comparison to each other under the results of the Wilcoxon signed-rank analysis.

A simple forecasting algorithm to predict wind energy using LSTM and DWT is proposed in [30]. The LSTM model is used to explore the non-linear behavior of wind power with time-sequenced data values. The non-stationarity of collected data over time is separated into elements with improved regularity and dependability using the wavelet packet approach. When RESs are employed increasingly, wind power prediction allows a different potentiality to magnify the security and solidity of the power channeling systems. In [31], a distinctive mixed forecast technique was developed for the prediction of basic electricity load and pricing. The learning strategy employs an LSTM network and a supervised neural method to improve prediction performance. In [32], utilizing the demand and temporal data over the period from Pennsylvania, New Jersey, Maryland power market, and Spain power market, the efficacy of the proposed strategy has been evaluated. For wind power uncertain interval prediction, a nonparametric SVM approach with controlled bandwidth is proposed [33]. Also, the K-Means LSTM neural network technique is employed for power forecasting [34]. The results demonstrate that the K-Means LSTM system model outperforms classification accuracy. Finally, the short-term wind energy is calculated using three deep-learning algorithms. Machine learning methods are presented for precise predictions of wind power [26].

The models mentioned in literature have contributed to the master time series, our proposed models have combined the power of CNN and LSTM with the optimizing capability of COA. We used COA as we must tune many parameters for accurate predictions of the model, to get rid of that problem we used the recently proposed optimization algorithm. The pictorial abstract of the suggested model is given in Fig. 1.

1.2. Research contributions

Besides the use of unconventional optimization methods like the Coati Optimization Algorithm (COA), this research describes a comprehensive approach for predicting the production of clean energy utilizing neural network (NN) algorithms, which are Long Short-Term Memory (LSTM) and Convolutional Neural Network (CNN). The input attributes used in the proposed modeling technique for solar are solar irradiance, normal irradiance, horizontal irradiance, air temperature, and atmosphere (atm.). The variables used in wind datasets are speed, direction, air temperature and pressure, humidity, and hypothetically available peak power output. The contributions of this work are summarized as follows.

1. For the projection of mixed solar-wind energy output, a hybrid Deep Neural Network (DNN)-based framework, (CNN-LSTM), is developed and discussed.
2. Complete data filtering is used to remove outliers from datasets and undesired data values that may influence the performance of the proposed technique. To measure the suggested structure's capacity (reduce the influence of outliers), the performance of proposed models employing datasets is thoroughly compared.
3. The effectiveness of the five forecasting models is evaluated using various subsets of data. Fig. 1 shows the models and generalized stepwise method: Preparing the data for extraction, extracting the features, training, and testing to minimize outliers, and analyzing the performance.
4. A unique COA is combined with CNN-LSTM (COA-CNN-LSTM) to update and fine-tune the model structure and hyperparameters because there is no straightforward theory for constructing and modifying the hyperparameters of the system.

The rest of the paper is organized as follows. Section 2 presents neural network models, including the structures of CNN and LSTM techniques, the hybrid CNN-LSTM approach, and COA. Section 3 describes the new COA-CNN-LSTM prediction model where the COA optimization technique is added to the hybrid approach (CNN-LSTM). Moreover, the objective function is described, and evaluation indicators are discussed. Then, the results and prediction models for hybrid PV/wind power are investigated in detail. In Section 4, observations about the research are examined. The conclusions are given in Section 5.

2. Proposed methods

The class of RNNs that LSTM belongs to is the most effective approach for times series or sequential data. LSTM is one of the most sophisticated versions of Gated Recurrent Units (GRUs). Whereas CNN-two dimensional (2D) is usually used to detect the hidden patterns in images and maximize the importance of feasible attributes or information. Later CNN-1D emerged as one of the promising techniques to get the relative features out of 1D data. Here we propose a model where we can get the benefits of combining both powerful techniques, LSTM and CNN algorithms. Thus, get the maximum advantage out of the available data. CNN is used to fetch beneficial information out of data. LSTM is used to remember the data sequence or to keep only the data important enough to obtain good results and accuracy and discard the data points that cannot generate the required results. In addition to the hybrid approach, we add an optimization technique (COA), which can converge to global minima with minimal hyper-tuning. Hyper parameter tuning is one of the main drawbacks of using deep learning algorithms. Therefore, COA discards this problem and converges to the global minima efficiently and timely [35]. Several neural network models are explored in this section. In this work, CNN-LSTM models are improved utilizing this brand-new method.

2.1. Long Short-Term Memory (LSTM) neural network

The assumption that all characteristics are unrelated to one another is one of the primary drawbacks of employing a regular neural network. However, in time series scenarios, it is typically not a sensible decision to make this hypothesis. On the other hand, RNNs offer different varieties of modern algorithms that equally perform well on sequential or time series data that depends on past observations. The motivation behind developing LSTM was to remove the vanishing gradients problem, which occurred with traditional RNNs when processing long-term or short-term dependent variables.

LSTM is one of the most promising algorithms that outperform all other algorithms in regard to sequential data with long-term or short-term dependencies due to its adaptability. LSTM comes under the umbrella of RNN (Fig. 2), which can be efficiently used by considering the sequential temporal data as input and output vectors into NN structure [36]. LSTM module consists of four structures rather than a single layer for an RNN module. Consequently, the input vector walks through NN cells in a successive form, output vectors are blended with subsequent continuous time input data at each stride, and the contents of the cell

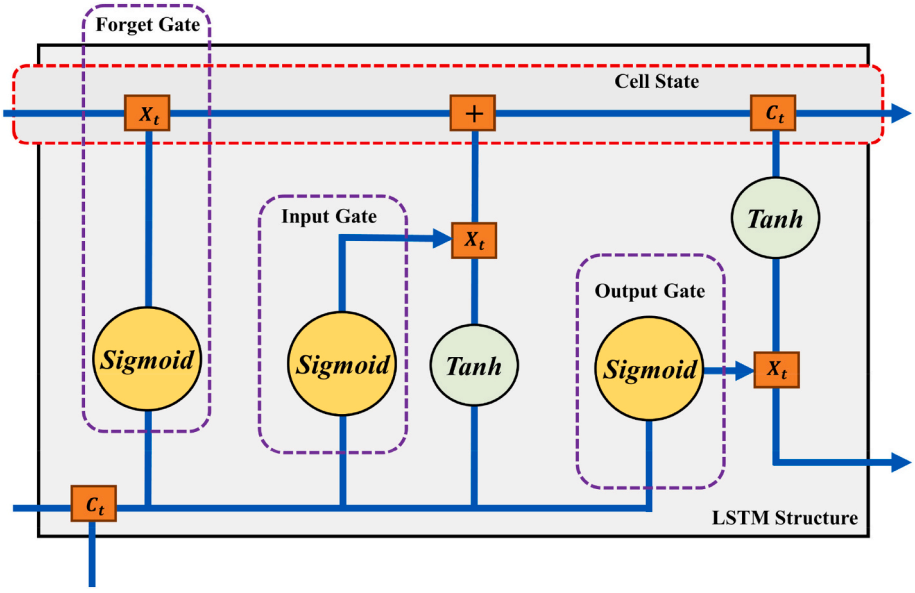


Fig. 3. LSTM Structure.

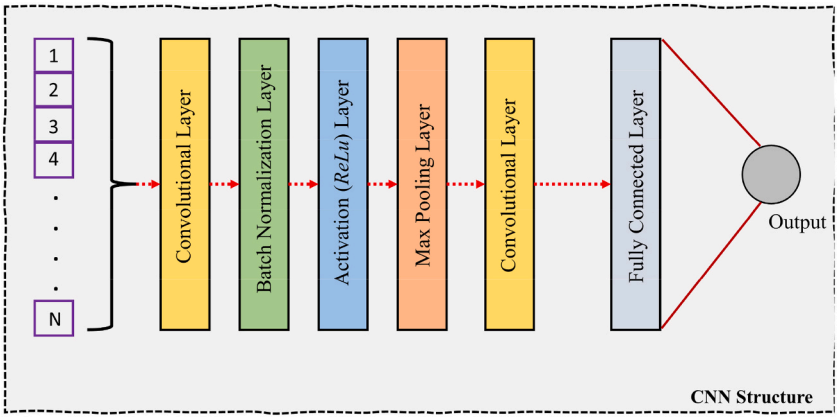


Fig. 4. CNN model architecture.

outcome are used as parameters in the successive iteration.

The main characteristic of an LSTM technique is the preservation of the cell state from the previous input pattern. Also, an important variable to consider in LSTM is that it can draw out the redundant information available in the input data vector. These prominent features can be utilized via an input gate, cell state (C_n), a forget gate (f_n), and an output gate (o_n). All these gates or layers interact with each other in a certain way. The fundamental structure of the LSTM model is displayed in Fig. 3.

Each one of these gateways takes on a state variable at a particular time phase. The calculated values are in the 0 to 1 forget gate range. Forget gate (f_n) helps to forget the redundant information, to save only the relative information Cn to proceed into prediction. For sequential modeling applications like text categorization and time series analysis, LSTM is an excellent tool [37–39]. An LSTM block's mathematical model is provided in Eqs. (1) to (6).

$$a_n\& = \sigma(x_n U^a + h_{n-1} W^a + b_a) \tag{1}$$

$$f_n\& = \sigma\left(x_n U^f + h_{n-1} W^f + b_f\right) \tag{2}$$

$$o_n\& = \sigma\left(x_n U^0 + h_{n-1} W^0 + b_o\right) \tag{3}$$

$$C_n\& = \sigma\left(f_n \odot C_{n-1} + a_n \odot \tilde{C}_n U^a + b_c\right) \tag{4}$$

$$\tilde{C}\& = tanh(x_n U^g + h_{n-1} W^g) \tag{5}$$

$$h_n\& = tanh(C_n) \odot o_n \tag{6}$$

where x_n represents the input matrix in the model network, h_n is the output value for a hidden layer, and C is a memory cell that makes it easier to remember a significant state that can be estimated using the mathematical formulas listed above. The old version of cell state Cn-1 is then updated to the new cell state Cn. New candidate values $\tilde{C}$ and output of the current LSTM block h_n employs the hyperbolic tangent function as in Eqs. (4) and (5). The core parameters of the neural networks, such as weights (W^a, W^f, W^o, W^g) and (U^a, U^f, U^o, U^g), as well as bias vectors (b_a, b_f, b_o, b_g), are the learnable parameters. The model updates the values associated with weights and biases by reducing the objective function. Multiplication of elements is indicated by the operator $\odot$. The sign sigma (σ) projects a sigmoid activation function, which is given by (7). *tanh* is the tan hyperbolic tangent function and is calculated by (8).

$$\sigma(X) = 1/(1 - e^{-z}) \tag{7}$$

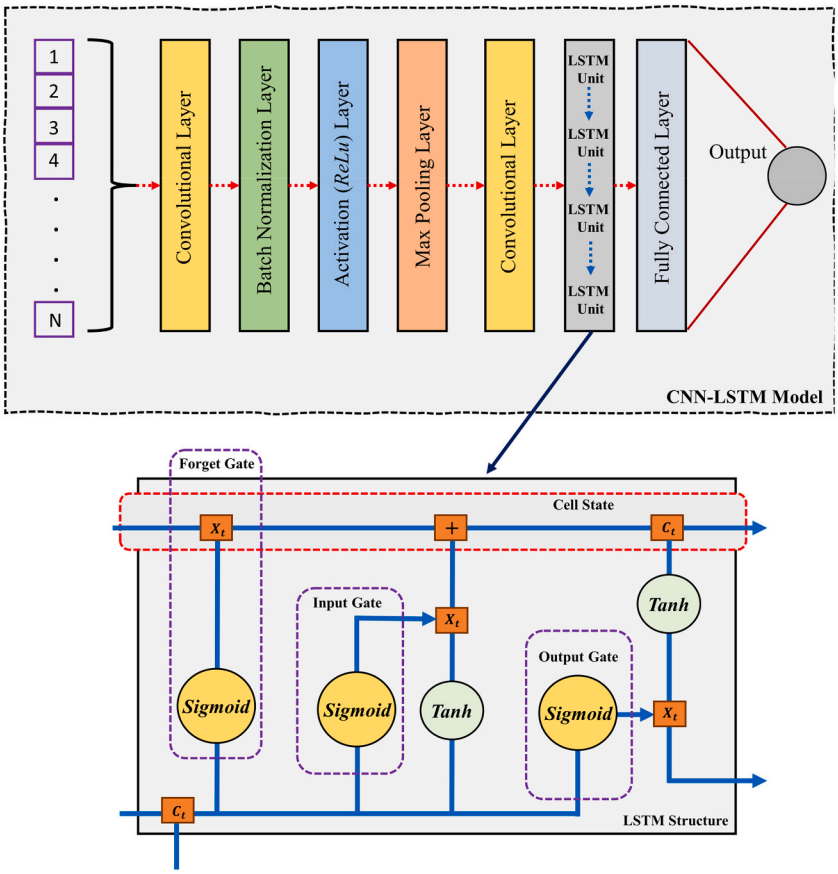


Fig. 5. CNN-LSTM Model.

$$tanh(X) = (e^{a} - e^{-a})/(e^{a} + e^{-a}) \tag{8}$$

2.2. Convolutional Neural Network (CNN)

An adjusted design of 2D-CNNs formulated as 1D-CNNs flourished in the near past. 1-D CNN is one of the common deep learning algorithms, usually an artificial neural feed-forward network composed of convolutional and pooling layers, as shown in Fig. 4.

To process the input through layers, neurons, and activation functions, the Rectified Linear Unit (ReLU) is typically utilized. This kind of activation is usually used to introduce non-linearity. Generally, Dropout layers and normalization (Scaling Data) techniques are used to avoid overfitting. Under similar conditions (architecture, network, and the number of hyper parameters), a 1D-CNNs' computational complexity is significantly lower than a 2D-CNNs'. Most 1D-CNN applications have often used small comprise of 1–2 hidden CNN layers architectures with networks having 50 or fewer units [40]. 1D-CNNs in their compact form are highly suited for low-cost and real-time applications [41], especially on handheld or mobile devices, because of their nominal proceeding needs. Compact 1D-CNNs have demonstrated better performance in the abovementioned neoteric research for applications like tagged data and significant signal changes from a variety of sources like ECGs, high-power circuitry, and time-bound predictions. The mean difference between 2D-CNN and 1D-CNN is that the latter model has 1D arrays to use as input vectors instead of matrices utilized in 2D-CNNs. Kernel and filter size are also replaced with 1D values. Consequently, 1D-CNN models process the raw data and extract useful information using a few layers and neurons available in architecture, as given in Eqs. (9) and (10).

$$a^{l}_{o,fl} = f\left(\sum_{\text{im}} a_{\text{i}}^{1-1} * k^{1}_{io,fl} + b^{1}\right) \tag{9}$$

$$a^{l}_{o} = f\left[max\left(\sum_{im} a_{i}^{l-1}\right) + b^{l}\right] \tag{10}$$

$$a^{l}_{o} = f\left(a_{i}^{l-1} * z^{l}_{io} + b^{l}\right) \tag{11}$$

where a is the 1D matrix ($n \times 1$) at input, f (.) represents the activation function, $k_{lio,fl}$ is mapping towards the kernel filter, which has dimensions of (k $\times$ 1), f tells the tally of filters being used, $a^{l}_{o,fl}$ is the output of lth convolutional layer, b denotes bias vectors, and z is the learnable parameter. As a result, the processes of feature extraction, regression, or classification are combined into a single process that may be adjusted to increase the performance of the classification or regression problem. As the only significant operation is a sequence of 1D convolutions (linear weighted summing of two 1D arrays), this is the main benefit of 1D-CNNs, which can lead to low computational cost. The linear operation can run simultaneously through the forward and back-propagation processes. This is an adjustable approach since the CNN architecture allows the output layer's sub-sampling factor to be adjusted to consider changes in the input layer's dimension. Fig. 5 shows the combination of the powerful techniques, LSTM and CNN algorithms.

2.3. Coati optimization algorithm (COA)

The problems in which there is a possibility of having multiple viable solutions are considered optimization problems, and the task of getting

the optimized combination of single or multi objective is considered an optimization task [42]. Optimization problems usually comprise three main features named decision variables, objective function, and constraints [43].

Coatis consume invertebrates like tarantulas and little vertebrate prey, such as small birds, rodents, lizards, eggs, and green iguanas. The iguanas are usually found in the deep forest or trees, so coatis chase them in batches. Some of these coatis climb up the tree and terrify the iguana into jumping to the ground while the other coatis wait on the ground to hunt them. Preying and avoiding predators can be modeled mathematically as an optimization process [44]. And the collective behavior of prey gives rise to swarming behavior, a standard trademark of stochastic optimization that helps to enhance the hunting and food-searching algorithms.

A. *Initialization*

Researchers have gained inspiration from the intelligent behavior of coatis, like attacking iguanas and confronting and running off from predators. The metaheuristic technique (COA) is a populace-based optimization algorithm in which they are observed as populace participants for this algorithm. The choice factor values depend on where each coati is in the total search space. Hence, Coatis' viewpoint presents a viable solution to the problems with the COA. It can be given mathematically as follows.

$$A_x : a_{x,y} = lb_y + r\cdot\left(ub_y - lb_y\right), x = 1, 2, \ldots, N, y = 1, 2, \ldots, m, \tag{12}$$

where A_x is the location of xth coati in a lookup field, $a_{x,y}$ is the estimate of the yth decision variable, N gives the information about the total number of coatis, and m and r represent the number of decision variables and randomly generated real numbers (ranging from 0 to 1), respectively. The terms (lb_y) and (ub_y) are the lower and upper bound values of the decision variable (y^{th}), respectively. The following matrix K given in Eq. (13), often known as the population matrix, serves as a mathematical representation of the coati-based populace.

$$K = \begin{bmatrix} A_1 \\ \vdots \\ A_i \\ \vdots \\ A_N \end{bmatrix}_{N\times m} = \begin{bmatrix} a_{1,1} & \cdots & a_{1,y} & \cdots & a_{1,m} \\ \vdots & \ddots & \vdots & . & \vdots \\ a_{x,1} & \cdots & a_{x,y} & \cdots & a_{x,m} \\ \vdots & . & \vdots & \ddots & \vdots \\ a_{N,1} & \cdots & a_{N,y} & \cdots & a_{N,m} \end{bmatrix}_{N\times m} \tag{13}$$

Several values for the problem's objective function are evaluated because of the placement of potential solutions in choice variables. These terms can be calculated by (14).

$$F = \begin{bmatrix} F_1 \\ \vdots \\ F_x \\ \vdots \\ F_N \end{bmatrix}_{N\times 1} = \begin{bmatrix} F(A_1) \\ \vdots \\ F(A_x) \\ \vdots \\ F(A_N) \end{bmatrix}_{N\times 1} \tag{14}$$

where F is the vector of the acquired objective function and F_x represents the value for the objective function generated using the xth coati iterations. The value of the objective function serves as a benchmark for candidate solution quality in evolutionary algorithms like the one presented for COA. A broad outline of this optimization method is provided. Candidates with the best results for the objective function are referred to as the most optimal populace member. The best individual in the sample is modified in each algorithmic iteration since the candidate solutions are improved throughout. In metaheuristic algorithms like COA, the value of the objective function acts as the standard for evaluating the quality of potential solutions. The placements of coatis' candidate solutions are often updated because of two natural actions of coatis, (a) a coati's attack method against iguanas and (b) an escape maneuver against predators. Also, Coatis updates its member population in two stages.

B. *Phase 1- Exploration Phase*

In their earliest phase, the attacking strategy is explored. The coatis' occupants are updated in the lookup field. According to this exploration phase, a batch of coatis jumps up to the tree to reach out for the iguana and attacks them while other coatis wait for them under the tree. It gives Coatis the power to move forward to various points in the global search space, simulating COA's capacity for exploration in the search problem-solving area. The iguana is in the finest location in this architecture for the best member of the community. Another assumption can be if coatis split into two equally populated batches. One batch climbs the tree to scare the iguana, and another seeks for the iguana to come tumbling down. The location for the coatis climbing the tree can be expressed as given in (15).

$$A_x^{P1} : a_{x,y}^{P1} = a_{x,y} + r\cdot\left(\text{Iguana}_y - I\cdot a_{x,y}\right), \text{for} x = 1,2,\ldots,\left\lfloor\frac{N}{2}\right\rfloor \text{ and } y = 1,2,\ldots,m. \tag{15}$$

The iguana is assumed to be in a random location inside the search area after it hits the ground. Considering this randomization behavior, coatis on the ground migrate, according to (16).

$$\begin{aligned} &\text{Iguana}^G : \text{Iguana}_y^G = lb_y + r\cdot\left(ub_y - lb_y\right), y = 1, 2, \ldots, m, \\ &A_x^{P1} : a_{x,y}^{P1} = \begin{cases} a_{x,y} + r\cdot\left(\text{Iguana}_y^G - I\cdot a_{x,y}\right), & F_{\text{Iguana}^G} < F_x, \\ a_{x,y} + r\cdot\left(a_{x,y} - \text{Iguana}_y^G\right), & \text{else}, \end{cases} \\ &\text{for } x = \left\lfloor\frac{N}{2}\right\rfloor + 1, \left\lfloor\frac{N}{2}\right\rfloor + 2, \ldots, N \text{ and } y = 1, 2, \ldots, m. \end{aligned} \tag{16}$$

For the updates, the newly estimated location for each coati is fair only if affecting the results of the objective function positively, or else the coati remains in the previous position. This upgraded case simulates values for $x = 1, 2\ldots, N$ and can be calculated by (17).

$$A_x = \begin{cases} A_x^{P1}, & F_x^{P1} < F_x \\ A_x, & \text{else} \end{cases} \tag{17}$$

where A_x^{p1} is the newly calculated position for the xth member of coatis, $A_{x,y}^{p1}$ is its yth dimension, F_x^{p1} calculate the objective function value and r randomly generate real numbers that range in [0,1], respectively. The finest possible position for the best member of Coatis is where the iguana is in the search space. Iguana_y is its yth dimension where I is an integer, which is selected randomly using the values available in the set [26]. Also, Iguana_G is the randomly generated location of the iguana on the ground, Iguana_{Gy} is its y^{th}dimension, $F_{\text{Iguana}}G$ is the objective function's value, and the symbol $\lfloor\cdot\rfloor$ represents the floor function (additionally referred to as the greatest integer function).

C. *Phase 2- Exploitation Phase*

The latter part of updating the location of coatis is inspired by coatis' escape from the attacks of predators and can be described mathematically considering this natural behavior of coatis. A coati retreats from its place when a predator pursues it. This technique places coati in a secure position near its present location. A random point is established close to the location of each coati to imitate as given by (18).

$$lb_y^{\text{local}} = \frac{lb_y}{n}, ub_y^{\text{local}} = \frac{ub_y}{n}, \text{where } n = 1, 2, \ldots, T.$$
$$A_x^{P2} : a_{x,y}^{P2} = a_{x,y} + (1 - 2r)\cdot\left(lb_y^{\text{local}} + r\cdot\left(ub_y^{\text{local}} - lb_y^{\text{local}}\right)\right), x = 1, 2, \ldots, N, y = 1, 2, \ldots, m \quad (18)$$

This is acceptable only if the newly calculated value improves the result of the provided goal metric which is given by Eq. (19)

$$A_x = \begin{cases} A_x^{P2}, & F_x^{P2} < F_x \\ A_x, & \text{else} \end{cases} \quad (19)$$

where A_x^{P2} is the recently calculated location for x^{th} coati, based on the lateral stage of COA, $a_{x,y}^{P2}$ is its y^{th}dimension, F_x^{P2} is the objective function value, r is a random number in the interval [0, 1], n is the iteration counter, lb_y^{local} and ub_y^{local} are the local lower upper bounds of the jth decision variable, respectively. lb_y and ub_y are the lower and upper bounds of the y^{th}decision variable, respectively.

3. Experimental test results

The COA-CNN-LSTM prediction models are provided in this section. The initial stages of data preparation and collecting are outlined. The objective function is then described. In addition, the evaluation indicators that are utilized to compare performance are studied one after another. The results and predictions model for hybrid PV/wind power are then reviewed. Table 1 provides detailed information about the wind farm installations, including nominal capacity, detailed turbine information like models and hub heights, and number of turbines. In addition, Table 2 presents PV installation information, including nominal power and other available information.

Table 1
Information about the Wind farms.

Wind Farm	Nominal Generation O/P Capacity (MW)	Wind Turbine Model	Detailed Turbine Information	Number of Turbines
Farm Site 1	75	GW1500/85	Capacity: 1500 kW Hub Height: 85.0 m Rotor Diameter: 87.0 m Website (07–2023): https://en.wind-turbine-models.com/turbines/1201-goldwind-gw-87-1500	50
	24	H93 L-2.000	Capacity: 2000 kW Hub Height: 85.5 m Rotor Diameter: 93.0 m Website (07–2023): https://market.hzwindpower.com/?Service/Pro/Product24/2.html	12
Farm Site 2	49.5	UP86–1500	Capacity: 1500 kW Hub Height: 80.0 m Rotor Diameter: 86.0 m Website (07–2023): https://en.wind-turbine-models.com/turbines/292-united-power-up1500-86	33
	49.5	UP82-1500	Capacity: 1500 kW Hub Height: 80.0 m Rotor Diameter: 82.0 m Website (07–2023): https://en.wind-turbine-models.com/turbines/292-united-power-up1500-86	33
Farm Site 3	201	GW3000/100	Capacity: 3000 kW Hub Height: 120.0 m Rotor Diameter: 140.0 m Website (07–2023): https://en.wind-turbine-models.com/turbines/1738-goldwind-gw-140-3000	67
Farm Site 4	30	FD89A-1500	Capacity: 1500 kW Hub Height: 85.0 m Rotor Diameter: 89.0 m Website (07–2023): https://en.wind-turbine-models.com/turbines/2224-dongfang-fd89-1500-geared	20
	36	FD116A-2000	Capacity: 2000 kW Hub Height: 90.0 m	18

Table 2
Information about the solar plants.

Solar Station Name	Nominal Generation Output Capacity (MW)	PV Panel Model	Manufacturer and Product Website	No. of PV Panels Installation
Solar Site 1	50	N/A	N/A	N/A
Solar Site 2	130	N/A	N/A	N/A
Solar Site 3	30	CS6U-325P	MFR: Canadian Solar Inc. *https://cn.csisolar.com/module/*	27,995
Solar Site 4	130	N/A	N/A	N/A

Table 3
Data statistics for four solar sites.

	Statistical Measures	Output Power (MW)	Solar Irradiance	Direct Normal Irradiance	Global Horizontal Irradiance	Air Temp.	Atm. pressure (hPa)
Solar Site 1	Mean	9.669	266.2	93.25	67.69	13.14	913.367
	Min.	0.000	0.000	0.000	0.000	−18.2	894.0
	Max.	48.32	1359	980.0	989.0	41.20	936.3
	Standard Dev.	13.70	367.89	200.78	111.98	14.37	8.740
Solar Site 2	Mean	19.56	169.3	122.154	78.30	13.70	861.03
	Min.	0.000	0.000	0.000	0.000	−13.92	844.51
	Max.	109.36	1041.93	751.75	561.80	40.47	936.3
	Std. Dev.	27.94	248.078	178.99	117.59	12.07	6.140
Solar Site 3	Mean	5.480	182.82	100.734	69.30	NA	1016.03
	Min.	0.000	0.000	0.000	0.000	NA	994.80
	Max.	29.11	1117.0	760.00	656.0	NA	1038.60
	Std. Dev.	8.234	294.58	185.03	101.88	NA	9.3230
Solar Site 4	Mean	19.46	150.15	139.51	20.84	18.71	1011.37
	Min.	0.440	0.000	0.000	0.000	−5.32	928.59
	Max.	114.68	1237.0	1010.27	150.96	49.8	1100.3
	Std. Dev.	27.45	253.43	210.68	31.48	10.27	33.220

3.1. Data collection and processing

In this paper, we assessed the proposed model (COA-CNN-LSTM) by utilizing the datasets generated on solar and wind-related features, and those datasets are collected by the Chinese State Grid 2021. The climate and electricity production are monitored every 15 min in live time. There is a description of the data collection and refining of data values. Using this information seems to hold promise for developing data-directed forecasting techniques for clean energy generation and enhancing electrical demand response (DR) programs for the power grid. Also, the mean, standard deviation, maximum, and lowest statistical descriptions of the solar and wind data values for the four sites are given in Table 3 and Table 4.

3.2. Objective function and evaluation parameters

For the training and validation of the proposed technique, an objective function (fitness function) is employed. The utility function's lesser values show how accurately the model's data predictions correspond to reality. Thus, the prediction accuracy is determined by the fitness function. The mean square error (MSE) is the most used fitness function and can be expressed in Eq. (20).

$$F.F = \frac{1}{n}\sum_{i=1}^{n}(T_i - P_i)^2, \tag{20}$$

where T_i, P_i and n represent the true values, anticipated values, and the tally of samples, respectively. Several error indices are also used to evaluate various models. Normalized root means square error (NRMSE) can be used to confirm the degree of findings' dispersion, as presented in (21). Mean absolute error (MAE) and mean absolute percentage error (MAPE) are indicators of forecast deviation, which are given in (22) and (23), respectively. Lastly, the R-squared value may be used to determine the correlation between actual and forecasted values, as shown in (24).

$$NRMSE = \frac{1}{T}\sqrt{\frac{1}{M}\sum_{a=1}^{M}(T_a - P_a)^2} \times 100\% \tag{21}$$

$$MAPE = \frac{1}{M}\sum_{a=1}^{M}\frac{|T_a - P_a|}{P_a} \times 100\% \tag{22}$$

$$NMAE = \frac{1}{M}\sum_{a=1}^{M}\frac{|T_a - P_a|}{P_a} \tag{23}$$

$$R^2 = \frac{\sum_{a=1}^{M}\left(T_a - \bar{T_a}\right) \bullet (P_a - \bar{P_a})^2}{\sum_{a=1}^{M}(T_a - \bar{T_a}) \bullet \sum_{a=1}^{M}(P_a - \bar{P_a})} \tag{24}$$

where, $\bar{T_a}$ is the average value of true output, and $\bar{P_a}$ represents the

Table 4
Data statistics for four wind sites.

	Statistical measures	Output Power (MW)	Wind speed @ height 10. (m/s)	Wind direction @ height 10. (°)	Wind speed @ height 30. (m/s)	Wind direction @ height 30. (°)	Air temp. (°C)	Atm. pressure (hPa)	Relative Humidity (%)
Wind Site 1	Mean	23.43	5.71	222.83	6.04	220.63	8.543	889.5	37.58
	Min.	0.00	0.00	0.00	0.00	0.00	−24.31	854.4	1.502
	Max.	98.1	25.47	25.46	29.19	359.09	36.13	918.19	93.12
	Std. Dev.	24.13	3.12	3.12	3.7	79.85	13.36	13.36	18.89
Wind Site 2	Mean	18.17	3.59	148.32	5.38	145.13	17.51	971.8	58.80
	Min.	0.67	0.00	0.00	0.00	0.003	−14.27	950.96	3.43
	Max.	94.27	27.93	360.0	22.09	360.0	36.32	990.73	100.0
	Std. Dev.	22.6	97.90	97.9	2.96	88.32	9.83	6.85	23.5
Wind Site 3	Mean	72.70	6.33	218.6	7.161	217.55	8.67	883.22	NA
	Min.	0.00	0.00	0.00	0.00	0.003	−24.547	441.27	NA
	Max.	201.35	23.30	359.92	27.12	360.0	37.59	900.45	NA
	Std. Dev.	55.74	4.021	85.71	4.49	88.46	13.23	7.900	NA
Wind Site 4	Mean	17.37	3.77	193.49	4.199	205.67	13.76	886.8	80.76
	Min.	0.00	0.00	0.00	0.00	0.00	−3.82	472.4	8.22
	Max.	64.61	17.74	356.6	20.4	357.0	35.25	902.0	100.0
	Std. Dev.	19.99	2.39	95.13	2.70	97.86	8.160	5.430	18.76

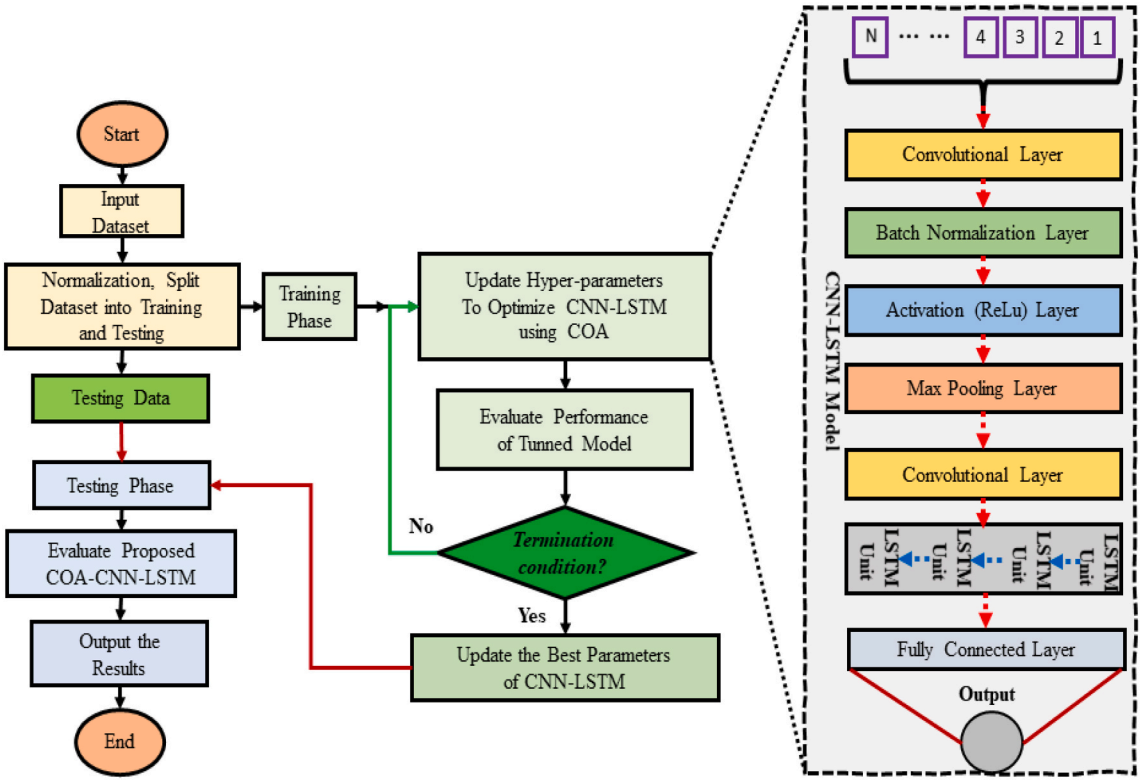


Fig. 6. Flow chart to represent the training using COA to optimize CNN-LSTM.

expected output's average value.

3.3. COA-based CNN-LSTM prediction model

Finding an optimizer that achieves a good balance between exploration and exploitation is important. Exploring the search space to find potential areas and then utilizing those regions to develop and refine the solutions are necessary for effective optimization. Deep learning models might contain many parameters. Therefore, the optimizer must be able to manage the high-dimensional search space properly. Additionally, to speed up the optimization process, consider if the optimizer supports parallel or distributed implementations if parallel computing resources are available. A good bio-inspired optimizer should include mechanisms that allow it to investigate several solutions effectively while converging on the best ones. Convergence speed considers the optimizer's convergence rate. Thus, some optimization techniques may converge to a satisfactory solution fast while others may require more rounds. Therefore, when utilizing sophisticated models or dealing with large-scale datasets, convergence speed will be essential.

In search of an optimizer, a fair trade-off between convergence speed and solution quality is considered. In addition, assess the optimizer's scalability concerning the problem size and the available computing power. Other issues are given as follows. 1) Robustness: the optimizer

Table 5
Hyperparameters for CNN-LSTM before applying COA.

Optimized Component	Hyperparameters	Range (To initialize)
Convolutional Layers	No. of Filters in 3 Layers	[20 – 29]
	Filter size in each Layer	[1 - 7]
	Activation	['LeakyReLU', 'ReLU', 'Tanh']
LSTM Layers	No. of Hidden Nodes/ Neurons	[10–500]
Learning Configuration	Learning Rate	[10-5 – 10-1]
	Dropout Rate	[0, 0.7]

Table 6
Information about data and implementation environment.

Feature	Value
Training Samples	80% of available data
Testing Samples	20 of remaining data samples
Vector Length	8
Sampling rate (min.)	15 min
Numerical Environment	Python 3.9.0
Libraries	Numpy, Scikit-Learn, TensorFlow, Pandas, Scipy
Machine Information	Intel(R) Core (TM) i5-7200U CPU @ 2.50GHz 2.71 GHz 8.00 GB RAM 64-bit operating system, x64-based processor

Table 7
RMSE, MEA, R^2, and RE results for four solar datasets using COA-CNN-LSTM and other models.

Error values	Site	COA-CNN-LSTM	PSO-CNN-LSTM	GWO-CNN-LSTM	CNN	LSTM
RMSE	Station 1	0.0387	0.0431	0.0391	0.0517	0.0411
	Station 2	0.0793	0.0811	0.0871	0.0896	0.0794
	Station 3	0.0249	0.0254	0.0251	0.0261	0.0255
	Station 4	**0.0140**	0.0194	0.0180	0.0240	0.0174
MAE	Station 1	0.0333	0.0391	0.0378	0.0481	0.0372
	Station 2	**0.0145**	0.0181	0.0196	0.0265	0.0216
	Station 3	0.0202	0.0287	0.0249	0.0319	0.0292
	Station 4	0.0174	0.0218	0.0199	0.0274	0.0239
R^2 score	Station 1	**0.9832**	0.9712	0.9669	0.9499	0.9571
	Station 2	0.9819	0.9719	0.9681	0.9506	0.9659
	Station 3	**0.9851**	0.9688	0.9694	0.9405	0.9633
	Station 4	0.9815	0.9676	0.9665	0.9519	0.9611
RE	Station 1	0.0042	0.2701	0.2991	0.1830	0.2079
	Station 2	**0.0013**	0.2013	0.2822	0.0691	0.1212
	Station 3	0.0694	2.4135	2.9189	2.1176	2.2815
	Station 4	0.0024	0.1189	0.1760	0.0961	0.1331

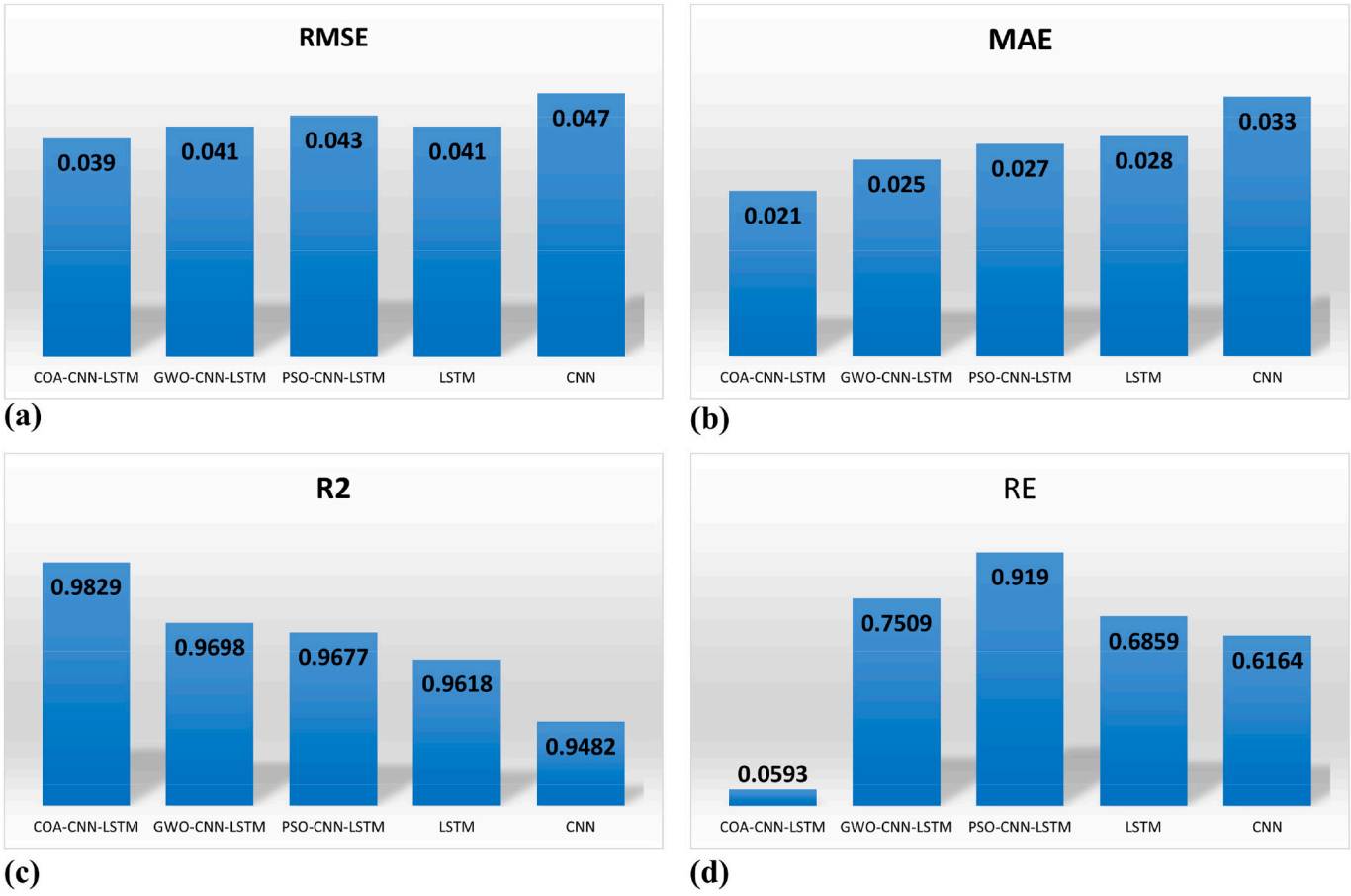


Fig. 7. Mean Errors for Solar Station Forecasting.

should behave robustly across many datasets and problem domains. It must deal with erratic or noisy fitness evaluations and prevent early convergence to poor choices. Robust optimization methods are adjusted to different aspects of the issue and deliver reliable performance. 2) Prior information incorporation: Depending on the details of the issue, limitations or prior information in the optimization procedure are included. Check whether the optimizer enables directing the search towards more helpful and relevant solutions by applying domain-specific information or restrictions during parameter adjustment. 3) Algorithm complexity and usability: Assess the optimization algorithm's complexity and usability in terms of implementation and parameter adjustment. While some optimizers have simple implementations and need a little setup, others could have more complicated algorithms or many parameters to adjust.

To generate the experimental results, we first identify the CNN-LSTM model's parameters that need to be improved. For example, architectural parameters (the number of layers and units), hyperparameters (learning rate and batch size), regularization parameters (dropout rate and weight decay), and any related parameters can be included. Then, create a fitness function that measures the CNN-LSTM model's performance for a specific parameter configuration. Any assessment parameter relevant to the issue area, including accuracy, error rate, and others, can serve as the foundation for the fitness function. The COA population's initial parameters are set to random or predetermined combinations within the specified parameter range. By training and analyzing the CNN-LSTM model with the appropriate parameter configuration, the fitness of each potential solution in the COA population is determined. Based on the model's performance as assessed by the fitness function, the fitness value is calculated. Following steps are carried out, as shown in Fig. 6.

i. Run the COA optimization procedure through iterations. Using COA optimization operators, such as search operators modeled from coati behavior, COA develops new candidate solutions with each cycle. The novel solutions indicate possible CNN-LSTM model parameter setups.
ii. Update the candidate solutions' fitness values per their performance during training and assessment using the fitness function. With this version, COA may direct the search to parameter settings that produce better performance.
iii. Establish convergence criteria to decide when the optimization process should end. Convergence may be determined by the following. A maximum number of iterations, a fitness improvement criterion, or other termination criteria.
iv. Get the Best Configuration: After optimization is complete, get the best parameter configuration that COA has identified. The CNN-LSTM optimized set of parameters is represented by this configuration.

3.4. Experimental results and discussion

To assess the accuracy of the predictions, a set of thorough assessment indicators is utilized using Eqs. (21) to (24). As mentioned above, the integration of COA serves purposes of robustness, reduced algorithms complexity, information incorporation, and high convergence speed. In this part, the experimental findings for the proposed COA and the comparative approaches are assessed. The model's accuracy is determined using all the statistical metrics. The generated results for several sites are presented below. It should be noted that the focus is on the key metrics that are frequently used to measure the precision of a regression model. The hyperparameters for all case studies are the same

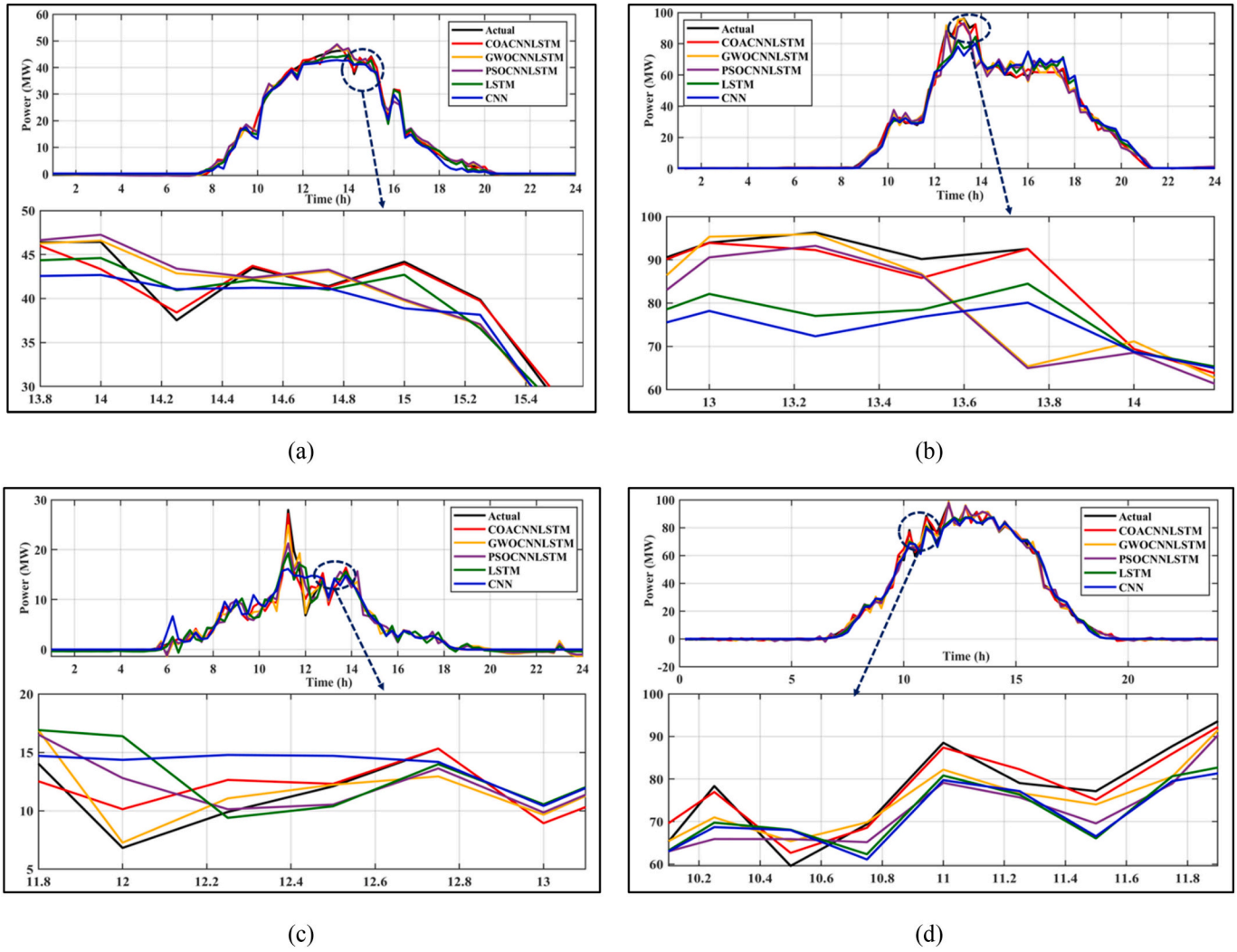


Fig. 8. Solar power station. (a) PV station 1. (b) PV station 2. (c) PV station 3. (d) PV station 4.

as provided in Table 5. In addition, the implementation environment parameters are given in Table 6.

3.4.1. Solar power forecasting

Table 7 gives the results of RMSE, MAE, R^2 score, and mean relative error for four solar datasets using the COA-CNN-LSTM and other comparative models, such as CNN, LSTM, and PSO-CNN-LSTM. Higher R^2 values and lower RMSE, MAE, and RE values indicate better performance. Fig. 7 shows the performance in terms of statistical data for solar power forecasting at four sites. Using the COA-CNN-LSTM method, the RMSE value is 0.039, as shown in Fig. 7(a). Compared to other models like CNN (0.047), the proposed model achieved a better score of 17.02%. As displayed in Fig. 7(b), the value of MAE is 0.021 (COA-CNN-LSTM). Compared to GWO-CNN-LSTM (0.025), PSO-CNN-LSTM (0.027), LSTM (0.028), and CNN (0.033), the performance was improved by 16%, 22.22%, 25%, and 36%, respectively. In addition, R^2 score of 0.9829 exhibits a higher degree of forecasting stability, as displayed in Fig. 7(c). The most significant improvement is achieved in RE, where the COA-CNN-LSTM model obtained a value of 0.0593 compared to the 0.919 by PSO-CNN-LSTM, as illustrated in Fig. 7(d). It is worth mentioning that the PSO-tuned parameters perform sub-optimally. Hence, the random parameters of stochastic optimization algorithms are not desired for trying CNN-LSTM models, and more sophisticated approaches yield more stable results.

In general, a model's performance is influenced by several variables, including the type and quantity of information incorporated into training and testing, the intricacy of the system's architecture, and its hyperparameters. The hyperparameters of the technique have been optimized. Then, based on its predictive accuracy, COA-CNN-LSTM performance can be evaluated. Fig. 8 displays a 24-h power estimation of different methods with a 15 min interval for four PV stations. From Fig. 8(a), a sample time between 2 and 4 pm is chosen for site 1. From Fig. 8(b), a sample time between 12 and 2 pm shows a large spike in available power. Usually, data fitting techniques fail to identify large fluctuations in short-term power forecasting, which cause glitches in power production. The model has been trained and validated on appropriate data. Using COA-CNN-LSTM for solar power prediction, the suggested technique can effectively handle characteristics present in data. COA-CNN-LSTM (in red) forecasted values are very close to the actual power (in black).

For station 3, displayed in Fig. 8(c), the RE for COA-CNN-LSTM, GWO-CNN-LSTM, PSO-CNN-LSTM, CNN, and LSTM are 0.069, 2.414, 2.919, 2.118, and 2.282, respectively. In addition, MAE follows similar trends and achieved values of 0.0202, 0.0287, 0.0249, 0.0319, and 0.0292, respectively. The R^2 values achieved by COA-CNN-LSTM are in the range of 0.9817–0.9845 showing strong correlations between forecasted power to the measured power. Fig. 8(d) shows improved performance and consistency. The zoomed-in part shows that the proposed COA-CNN-LSTM forecasts are close to the actual power.

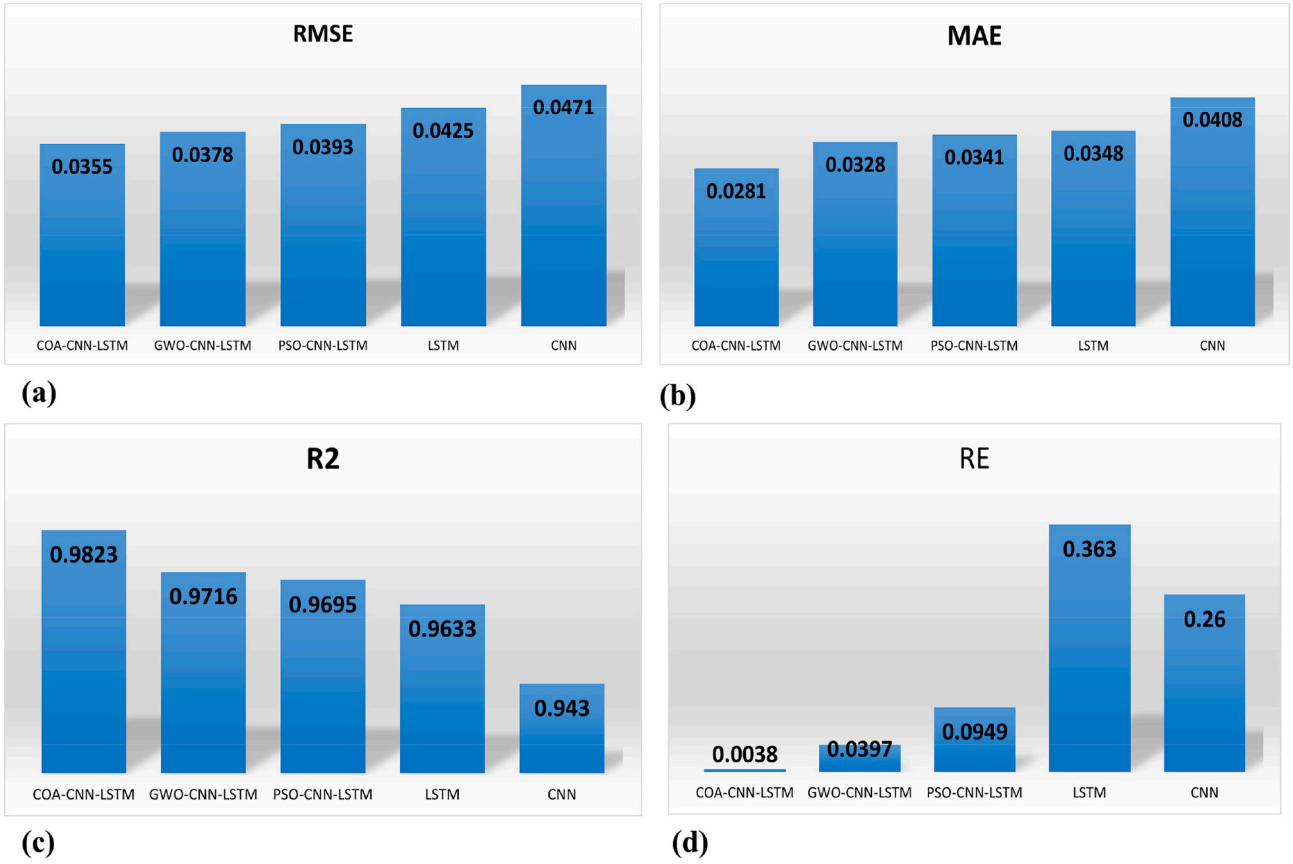


Fig. 9. Mean errors for wind station forecasting.

3.4.2. Wind power forecasting

The difference between wind power and solar power forecasting is that wind energy can be available 24/7, highly stochastic in nature, and wind speed has an exponentially increasing correlation with the output power of the wind energy conversion system (WECS). On the other hand, solar energy is available only in the daytime and follows restricted deviation, where peak hours are often between 10 am and 3 pm. Therefore, the output of solar panels is not as complicated as in WECS. Using the historical dataset of wind power and meteorological factors, as given in Table 4, the model is trained and validated. The performance of the COA-CNN-LSTM algorithm is evaluated as follows.

Fig. 9 shows the performance regarding statistical indices for wind power forecasting at 4 sites using RMSE, MAE, R^2, and RE. Table 8 provides a comprehensive summary of the results. RMSE values of site 1 for COA-CNN-LSTM, GWO-CNN-LSTM, PSO-CNN-LSTM, CNN, and LSTM are 0.0412, 0.045, 0.047, 0.057, and 0.05, respectively. Similarly, site 2, site 3, and site 4 results show 4.8% to 19.38% lower RMSE values. The RE and MAE values follow similar trends. The R^2 values achieved by COA-CNN-LSTM are in the range of 0.9817–0.9845, showing strong correlations between forecasted wind power to the measured one. Comparatively, the smallest R^2 value is achieved by the CNN model (0.9299), LSTM scored 0.9559, and PSO-CNN-LSTM has similar limitations and scored lower than 0.9669 due to the undesired random behavior of the PSO algorithm. Again, RE shows the highest differential. All comparative results assert the superior performance of the presented model.

Fig. 10 shows a 24-h power estimation of different methods with a 15 min interval for four wind farms in China. For site 1, the proposed model has performed better than CNN, LSTM, and PSO-CNN-LSTM models. Our study demonstrates the effectiveness of the COA-CNN-LSTM model for wind power forecasting and highlights its potential to increase the operational effectiveness and profitability of wind energy systems. For wind farm in China (site 3), the outcomes demonstrate that the COA-CNN-LSTM model outruns both the LSTM and the conventional CNN time series models. The COA-CNN-LSTM is found to be particularly useful in accurately predicting short-term variations in wind power generation, which is important for effective wind farm operation. The results of site 4 show that the COA-CNN-LSTM model performs well in predicting wind power. The model is found to be particularly useful for wind power forecasting at different time horizons, from short-term to long-term forecasting, and it can be used for wind power integration into the electricity grid.

Table 9 presents a summary of the COA-CNN-LSTM average performance capability for wind and solar power forecasting. It is concluded that the COA-CNN-LSTM model has shown promising results in solar and wind power forecasting and can be used as an effective tool for clean energy power plant operation and energy management.

4. Common observations

The nominal capacity of sites and power generation was given in Table 7. The 24-h power generation by the proposed model shows improvised forecasting stability. The confidence interval of forecasting increases significantly with more precise prediction [45]. The normalized RMSE is a metric used to measure the spread of errors in prediction, with a higher value indicating a wider spread. To ensure fairness, 20 runs are conducted for each analysis. However, it can generate unrealistic magnitude of forecasted power owing to faulty values of error margin when actual values should be zero or very small. The normalized MAE measures the average size of absolute errors between actual and predicted values, taking absolute values into account to prevent cancellation between negative and positive errors [46]. A low value of error indicates high prediction accuracy. The smaller the values of MAE, RMSE, and RE, the better the forecasting performance [47].

Table 8
RMSE, MEA, R^2, and RE results for four wind datasets using COA-CNN-LSTM and other models.

Error values	Site	COA-CNN- LSTM	GWO-CNN-LSTM	PSO-CNN-LSTM	CNN	LSTM
RMSE	Wind site 1	0.0411	0.0451	0.0472	0.0572	0.0501
	Wind site 2	0.0309	0.0311	0.0347	0.0413	0.0371
	Wind site 3	0.0426	0.0464	0.0449	0.0511	0.0487
	Wind site 4	**0.0274**	0.0289	0.0305	0.0388	0.0343
MAE	Wind site 1	0.0501	0.0551	0.0572	0.0612	0.0598
	Wind site 2	0.0311	0.0339	0.0367	0.0415	0.0371
	Wind site 3	**0.0104**	0.0161	0.0149	0.0220	0.0187
	Wind site 4	0.0211	0.0263	0.0277	0.0388	0.0239
R^2 score	Wind site 1	0.9832	0.9717	0.9669	0.9299	0.9592
	Wind site 2	0.9799	0.9622	0.9681	0.9411	0.9559
	Wind site 3	0.9817	0.9774	0.9714	0.9585	0.9683
	Wind site 4	**0.9845**	0.9751	0.9719	0.9425	0.9701
RE	Wind site 1	0.0074	0.0529	0.0919	0.1154	0.0400
	Wind site 2	**0.0021**	0.0778	0.1223	0.1813	0.1522
	Wind site 3	0.0051	0.0229	0.1281	0.5495	0.2966
	Wind site 4	0.0007	0.0052	0.0375	0.1953	0.9655

Compared to CNN and LSTM, the proposed COA-CNN-LSTM achieved 27.81% less RMSE, 0.3926 less MRE, 90% less MAE, and 0.1729–0.04332 more R^2 values. Thus, the prediction accuracy was improved. Short-term power prediction comparison validates COA-CNN-LSTM performance with higher accuracy. The investigational outcomes demonstrate that the proposed technique achieves higher forecasting accuracy measured to that of state-of-the-art models. Specifically, the proposed model achieves an MAE of 5.23% and an RMSE of 6.2% on the testing set, while the state-of-the-art models achieve an MAE of 7–10% and an RMSE of 7.9%–11%. The proposed model also exhibits a higher correlation coefficient and lower prediction error distribution compared to the SOTA models. These outcomes suggest that the proposed model can provide precise and viable short-term wind power prophecy for wind energy systems [48].

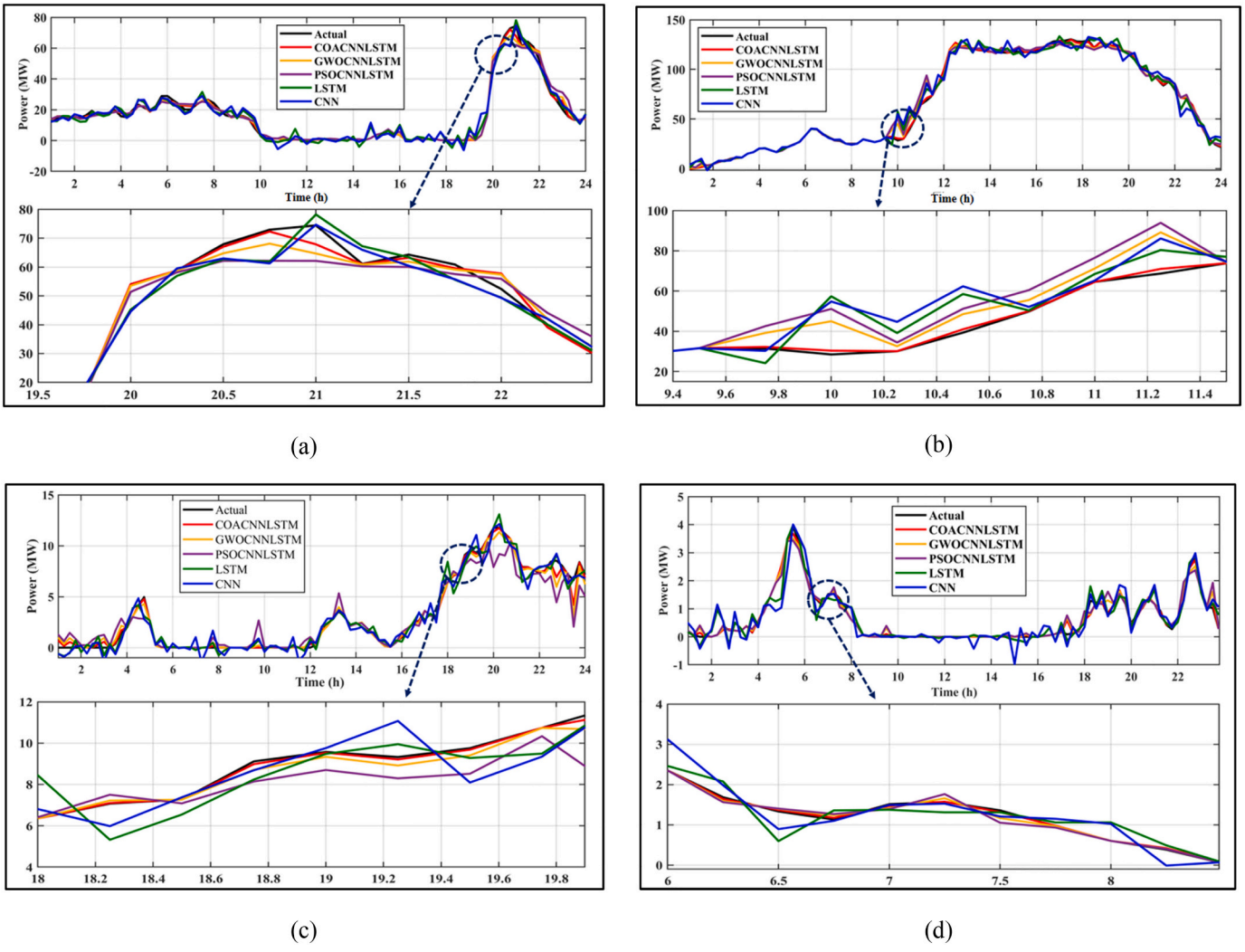


Fig. 10. Wind farms. (a) Wind station 1. (b) Wind station 2. (c) Wind station 3. (d) Wind station 4.

Table 9
COA-CNN-LSTM average performance capability summary for wind and solar power forecasting.

Site Wind/Solar	Nominal Capacity	Power generation forecast			
		RMSE	MAE	R^2	RE
Wind farm site 1	99 MW	0.0411	0.0501	0.9832	0.0074
Wind farm site 2	99 MW	0.0309	0.0311	0.9799	0.0021
Wind farm site 3	201 MW	0.0426	0.0104	0.9817	0.0051
Wind farm site 4	66 MW	0.0274	0.0211	0.9845	0.0007
Solar farm site 1	50 MW	0.0387	0.0333	0.9832	0.0042
Solar farm site 2	130 MW	0.0793	0.0145	0.9819	0.0013
Solar farm site 3	30 MW	0.0249	0.0202	0.9851	0.0694
Solar farm site 4	130 MW	0.0140	0.0174	0.9815	0.0024

4.1. Forecasting stability

The suggested methods provide good predictability and stability. The indicator of forecasting error performance variance. The stability of a prediction can be evaluated using low variability indices and the smallest absolute error magnitude, which indicate a higher level of accuracy. Established approaches for forecasting are less affected by outliers and produce predictions within 2 to 5% of the average of other methods. Multiple runs analysis confirms the stability of the proposed model COA-CNN-LSTM and is compared to PSO-CNN-LSTM and GWO-CNN-LSTM while CNN and LSTM models show unbalanced prediction performance [49].

4.2. Nash-Sutcliffe metric

The accuracy is assessed statistically using the Nash-Sutcliffe model. The Nash-Sutcliffe efficiency measures how well the model can forecast time series data [50]. The Nash-Sutcliffe constant (NSC) may vary between $-\infty$to 1. The values closer to 1 indicate perfect tracking or fit values. For example, for NSC magnitude higher than 0.75 refers to a good performance, between 0.36 and 0.75 is acceptable, and lower ranges refer to poor performance. The Nash-Sutcliffe coefficient is calculated by (25).

$$NSC = 1 - \frac{\sum_{i=1}^{n}(OV - PV_i)^2}{\sum_{i=1}^{n}(OV - OVM)^2} \tag{25}$$

where OV is the observed magnitude value, the ith predicted value is given by PV_i, and the OVM is the observed value mean for n number of samples. An r-square factor is produced for the NSC using a Coefficient value of 1, which denotes a perfect match between the seen and projected data. The model is not able to estimate any more than using the mean of the observed data when the r-square values are equal to or <0. Ockham's razor should be used as a result. To imitate empirical research studies, recorded data are combined using an intersection. One critical assumption of this measurement is the data are normal [51]. The average NSC values for COA-CNN-LSTM, GWO-CNN-LSTM, PSO-CNN-LSTM, CNN, and LSTM are 0.976, 0.97, 0.938, 0.932, and 0.945, respectively. It is also worth noticing that the case studies involving offshore data show higher NSC averages, because the wind speeds are more consistent offshore. Since the PSO generates more random behavior, the class score is higher for the land-based studies, as shown in Table 10.

4.3. Granger causality test

The Granger causality test is a popular statistical technique for projecting short-term wind output. It determines the causal relationship between wind speed and wind power, which develops accurate forecasting and reliable models. The test looks at whether previous values of one variable can be used to predict future values of another variable using time series data. The test determines whether wind speed is a leading indicator of wind power. If it is, the wind speed can be used to forecast wind power with higher accuracy. The Granger causality test is useful for short-term wind power forecasting because it considers the time lags between wind speed and wind power, as the fact that changes in wind speed may take some time to translate into changes in wind power. By analyzing the time lags between the two variables, the test can provide insights into the dynamics of the wind power system. Then, this information can be used to develop forecasting models that capture the complex relationship between wind speed and wind power.

The Granger causality test [52] identifies if a collection of m_1 time series variables, referred to as the "cause" variables, have changed over time and whether these changes have had an impact on the predicted distribution of a different set of m_2 time series variables, referred to as the "effect" variables. Data should be modified to avoid the risk of autocorrelation. The model is also checked for unit roots, which would bias the test results. Testing the null hypothesis and alternative hypothesis comes first. For example, y(t) does not Granger-cause x(t). Choose the delays in the subsequent step. To determine if the findings are the same for various lag levels, it might be simpler to just choose a few numbers and perform the Granger test numerous times. Lags should not affect the outcomes at all. Step three involves the calculation of the f-value using Eqs. (26) and (27), the value of β_j=0 is calculated for all lags (j) by (28). Step five calculates the F-statistics using (30).

$$y(t) = \sum_{i=1}^{\infty}\alpha_i y(t-i) + c_1 + v_1(t) \tag{26}$$

$$y(t) = \sum_{i=1}^{\infty}\alpha_i y(t-i) + \sum_{i=1}^{\infty}\beta_i y(t-j) + c_2 + v_2(t) \tag{27}$$

To check $y(t)$ Granger-causes $x(t)$

$$y(t) = \sum_{i=1}^{\infty}\alpha_i y(t-i) + c_1 + u_1(t) \tag{28}$$

$$y(t) = \sum_{i=1}^{\infty}\alpha_i y(t-i) + \sum_{i=1}^{\infty}\beta_i y(t-j) + c_2 + u_2(t) \tag{29}$$

$$F = \frac{(ESS_R - ESS_{UR})/q}{ESS_{UR}/(n-k)} \tag{30}$$

Neglect the null values if the F-statistic is higher than the f-value. According to R-Values, there are substantial correlations between wind

Table 11
Comparison of models based on GCT average performance.

Model	F-value	P value	T statistic	Error Standard (≤)
COA-CNN-LSTM	0.0992	0.018	0.901	0.10
GWO-CNN-LSTM	0.1097	0.032	0.781	0.10
PSO-CNN-LSTM	0.1005	0.046	0.782	0.12
CNN	0.1272	0.043	0.882	0.13
LSTM	0.9305	0.039	0.760	0.13

Table 10
NSC average comparisons for each site.

Model	Site: 1	Site: 2	Site: 3	Site: 4	Site: 5	Site: 6	Site: 7	Site: 8	Avg.
COA-CNN-LSTM	0.98	0.98	0.97	0.97	0.98	0.98	0.98	0.97	0.976
GWO-CNN-LSTM	0.98	0.97	0.97	0.97	0.96	0.97	0.98	0.96	0.970
PSO-CNN-LSTM	0.96	0.96	0.92	0.93	0.92	0.94	0.96	0.97	0.938
CNN	0.94	0.94	0.93	0.93	0.92	0.93	0.95	0.96	0.932
LSTM	0.95	0.94	0.95	0.95	0.94	0.94	0.94	0.95	0.945

Table 12
Comparison of SOTA models with the proposed technique.

Reference	Technique and description	Accuracy/ Nash-Sutcliffe metric	RMSE/MAE/RE	R^2 / correlation/ Direct accuracy (DA)	Runtime
[53]	Dual-stage self-attention GRU, DSSAM-GRU	0.9803	RMSE: 148.07 MAE: 88.4615	DA: 0.7872	329.65 s
[54]	ANN trained by Ant colony optimization: MD-ACO-2NN	0.9768	RMSE: 1.4512 NMSE: 0.1019	0.8981	NA
[55]	Federated Deep learning model DDPG-based prediction	0.9708	NMAE: 0.0206514 NRMSE: 0.0377521	NA	NA
[56]	Boosted ANFIS, MPAmu-ANFIS	0.915	RMSE: 0.175686 MAE: 0.005023 MARE: 0.000002	0.991540	NA
[26]	HBO-LSTM	0.965	RMSE: 0.045809 MAE: 0.032507 MARE: 0.786838	0.962618	664.13 s
This work	COA-CNN-LSTM	0.98	RMSE: 0.0140 MAE: 0.0174 RE: 0.0013	0.9832	32 s

speed, temperature, and humidity. Thus, the corresponding output power has a long-term stable relationship. Hence, the Granger causality test may be run.

The final step is to perform a hypothesis test to determine whether the independent variables have a statistically significant effect on the dependent variable. This is done by comparing the model with the independent variables to a null model without the independent variables. The null hypothesis is that the independent variables do not have a causal effect on the dependent variable. The basic form of the Granger causality test is:

H0: lagged values of X do not Granger-cause Y.

Ha: lagged values of X do Granger-cause $Y$where X is the predictor variable and Y is the target variable. Therefore, the statistical test is used to examine if one time series can forecast another. Let's assume two time series, (Y and X) with observations Y_t and X_t at time t. The model can be written as follows.

$$Y_t = \alpha_Y + \sum \beta_{Yi}\, Y_t - i + \sum \gamma_{Yi} X_{t-i} + \varepsilon_{Yt} \tag{31}$$

$$X_t = \alpha_X + \sum \beta_{Xi} X_t - i + \sum \gamma_{Xi}\, Y_{t-i} + \varepsilon_{Xt} \tag{32}$$

where α_Y and α_X are intercept terms, β_{Yi} and β_{Xi} are coefficients for the past values of Y and X, and γ_{Yi} and γ_{Xi} are coefficients for the past values of X and Y, respectively. ε_{Yt} and ε_{Xt} are error terms. To test for Granger causality, we first estimate the above model using suitable estimation techniques, such as ordinary least squares (OLS). Then the F-statistic for the anthropic principle is calculated so that the coefficients γ_{Yi} are all equal to 0 using (33).

$$F = \left[\frac{(RSS_R - RSS_U)}{m}\right] \Big/ \left[\frac{RSS_U}{(n - 2m - 1)}\right] \tag{33}$$

where RSS_R is the residual sum of squares for the restricted model (i.e., where $\gamma_{Yi} = 0$ for all i), RSS_U is the residual sum of squares for the unrestricted model, m is the number of restrictions like the number of γ_{Yi} coefficients, and n is the sample size. The null hypothesis is disregarded, and it is decided that X Granger causes Y and vice versa if the F-statistic exceeds a threshold value established according to the desired level of significance and the degrees of freedom. The results are summarized in Table 11. They indicate lower F- and P-values by the proposed COA-CNN-LSTM model with the highest T statistics of 0.901 and error margin of $\leq$0.1. It indicates that the forecasted results closely follow the original production.

4.4. Comparison against SOTA models

The results of the proposed model have been compared to the recently published SOTA models. A comparison summary with SOTA is given in Table 12. The analyses show that the COA-CNN-LSTM model is performing well in terms of some criteria like forecasting stability, statistical error, and correlation. This comparison is diversified because some studies may not utilize similar datasets, analysis tools, hardware resources, or statistical indices. Still, the comparison indicated the capability of the proposed model compared to the recent SOTA models and the redundancy of the results for experimental setups. The analysis showed that the DSSAM-GRU and HBO-LSTM models have very high runtimes since they require more computational time to generate higher accuracy.

5. Conclusion

The paper presented an effective method to deal with short-term wind and solar power forecasting. The COA-CNN-LSTM model has proven to be efficient at eight different sites. The model used Granger causality to identify causal relationships between wind power generation and meteorological variables and integrated this information into its forecasting process using convolutional and recurrent neural networks. The evaluation of the model using some metrics like RMSE and R^2 demonstrated its high accuracy and ability to capture complex temporal patterns in wind power generation data. The model also outperformed baseline models, indicating its superiority in capturing the underlying dynamics of wind power generation.

The findings of this study have important implications for the renewable energy industry, as accurate forecasting of wind and solar power generation can improve grid stability and facilitate the integration of clean energy into the utility grid. Compared to GWO-CNN-LSTM, LSTM, CNN, and PSO-CNN-LSTM models' short-term predictions RMSE were decreased by 0.5% and 5.8% for hour and day-ahead, respectively, with nMAE of 4.6%, RE 27% and nRMSE of 6.2%. The Granger causality test score is 0.0992 and the Nash Sutcliffe constant is 0.98. Overall, the proposed based model provides a promising approach for wind power prediction that might be applied to a range of different sites and conditions.

CRediT authorship contribution statement

Mohamad Abou Houran: Visualization, Validation, Supervision, Formal analysis, Funding acquisition, Methodology, Writing - original draft, Writing - review & editing. **Syed M. Salman Bukhari:** Validation, Investigation, Formal analysis, Data curation, Methodology, Writing - review & editing. **Muhammad Hamza Zafar:** Writing – review & editing, Visualization, Software, Methodology, Formal analysis. **Majad Mansoor:** Validation, Software, Formal analysis, Writing - review & editing. **Wenjie Chen:** Resources, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

[1] Mansoor M, et al. Novel grass hopper optimization based MPPT of PV systems for complex partial shading conditions. Solar Energy 2020;198:499–518.
[2] Rouani L, et al. Shading fault detection in a grid-connected PV system using vertices principal component analysis. Renew Energy 2021;164:1527–39.
[3] Kaldellis JK, Kavadias KA. Cost-benefit analysis of remote hybrid wind-diesel power stations: case study Aegean Sea islands. Energy Policy 2007;35(3):1525–38.
[4] Zafar MH, et al. Adaptive ML-based technique for renewable energy system power forecasting in hybrid PV-wind farms power conversion systems. Energ Conver Manage 2022;258:115564.
[5] Kong R, et al. Performance and economic evaluation of a photovoltaic/thermal (PV/T)-cascade heat pump for combined cooling, heat and power in tropical climate area. J Energy Storage 2020;30:101507.
[6] Tayab UB, et al. Solar photovoltaic power forecasting for microgrid energy management system using an ensemble forecasting strategy. Energy Sour Part A Recover Util Environ Eff 2022;44(4):10045–70.
[7] Kondaiah V, et al. A review on short-term load forecasting models for micro-grid application. J Eng 2022;2022(7):665–89.
[8] Zhao L, Zhang X, Peng X. Power fluctuation mitigation strategy for microgrids based on an LSTM-based power forecasting method. Appl Soft Comput 2022;127:109370.
[9] Sun MC, Feng C, Zhang J. Conditional aggregated probabilistic wind power forecasting based on spatio-temporal correlation. Appl Energy 2019;256:113842.
[10] Yongning, et al. Correlation-constrained and sparsity-controlled vector autoregressive model for spatio-temporal wind power forecasting. IEEE Trans Power Syst 2018;33(5):5029–40.
[11] Xie L, et al. Short-term spatio-temporal wind power forecast in robust look-ahead power system dispatch. IEEE Trans Smart Grid 2013;5(1):511–20.
[12] Abbas T, et al. Greener energy: issues and challenges for Pakistan-geothermal energy prospective. Renew Sustain Energy Rev 2014;31:258–69.
[13] Lu P, et al. A novel spatio-temporal wind power forecasting framework based on multi-output support vector machine and optimization strategy. J Clean Prod 2020;254:119993.
[14] Dong X, et al. Spatio-temporal convolutional network based power forecasting of multiple wind farms. J Mod Power Syst Clean Energy 2021;10(2):388–98.
[15] Mao MX, et al. Classification and summarization of solar photovoltaic MPPT techniques: a review based on traditional and intelligent control strategies. Energy Rep 2020;6:1312–27.
[16] Al-Shetwi AQ, et al. Grid-connected renewable energy sources: review of the recent integration requirements and control methods. J Clean Prod 2020;253:119831.
[17] Gayatri MTL, Parimi AM, Kumar AVP. A review of reactive power compensation techniques in microgrids. Renew Sustain Energy Rev 2018;81:1030–6.
[18] Mellit A, Kalogirou SA. MPPT-based artificial intelligence techniques for photovoltaic systems and its implementation into field programmable gate array chips: review of current status and future perspectives. Energy 2014;70:1–21.
[19] Hasan R, et al. Grid-connected isolated PV microinverters: a review. Renew Sustain Energy Rev 2017;67:1065–80.
[20] Champier D. Thermoelectric generators: a review of applications. Energ Conver Manage 2017;140:167–81.
[21] Rajesh R, Mabel MC. A comprehensive review of photovoltaic systems. Renew Sustain Energy Rev 2015;51:231–48.
[22] Ahmad T, Zhang DD, Huang C. Methodological framework for short-and medium-term energy, solar and wind power forecasting with stochastic-based machine learning approach to monetary and energy policy applications. Energy 2021;231:120911.
[23] Zhao XY, et al. Short-term probabilistic predictions of wind multi-parameter based on one-dimensional convolutional neural network with attention mechanism and multivariate copula distribution estimation. Energy 2021;234:121306.
[24] Yu RG, et al. LSTM-EFG for wind power forecasting based on sequential correlation features. Future Gener Comput Syst Int J Esci 2019;93:33–42.
[25] Liu J, et al. A hybrid GA-PSO-CNN model for ultra-short-term wind power forecasting. Energies 2021;14(20):6500.
[26] Ewees AA, et al. HBO-LSTM: optimized long short term memory with heap-based optimizer for wind power forecasting. Energ Conver Manage 2022;268:116022.
[27] Wang XC, Guo P, Huang XB. A review of wind power forecasting models. In: Proceedings of international conference on smart grid and clean energy technologies (Icsgce 2011). 12; 2011. p. 770–8.
[28] Mahmoud T, Dong ZY, Ma J. An advanced approach for optimal wind power generation prediction intervals by using self-adaptive evolutionary extreme learning machine. Renew Energy 2018;126:254–69.
[29] Shahid F, Zameer A, Muneeb M. A novel genetic LSTM model for wind power forecast. Energy 2021;223:120069.
[30] Zeng C, et al. Predicting vacant parking space availability: a DWT-bi-LSTM model. Phys a-Stat Mech Appl 2022;599:127498.
[31] Memarzadeh G, Keynia F. Short-term electricity load and price forecasting by a new optimal LSTM-NN based prediction algorithm. Electr Pow Syst Res 2021;192:106995.
[32] Heydari A, et al. Short-term electricity price and load forecasting in isolated power grids based on composite neural network and gravitational search optimization algorithm. Appl Energy 2020;277:115503.
[33] Zhou BW, et al. Wind power prediction based on LSTM networks and nonparametric kernel density estimation. Ieee Access 2019;7:165279–92.
[34] Dong WC, et al. Short-term wind-speed forecasting based on multiscale mathematical morphological decomposition, K-means clustering, and stacked denoising autoencoders. Ieee Access 2020;8:146901–14.
[35] Qu JQ, Qian Z, Pei Y. Day-ahead hourly photovoltaic power forecasting using attention-based CNN-LSTM neural network embedded with multiple relevant and target variables prediction pattern. Energy 2021;232:120996.
[36] Kim TY, Cho SB. Predicting residential energy consumption using CNN-LSTM neural networks. Energy 2019;182:72–81.
[37] Rick R, Berton L. Energy forecasting model based on CNN-LSTM-AE for many time series with unequal lengths. Eng Appl Artif Intel 2022;113:104998.
[38] Li K, et al. Ultra-short term power load forecasting based on CEEMDAN-SE and LSTM neural network. Energ Buildings 2023;279:112666.
[39] Mansoor M, et al. Hybrid forecasting models for wind-PV systems in diverse geographical locations: performance and power potential analysis. Energ Conver Manage 2023;287:117080.
[40] Ozcanli AK, Baysal M. Islanding detection in microgrid using deep learning based on 1D CNN and CNN-LSTM networks. Sustain Energy Grids Netw 2022;32:100839.
[41] Ahmed R, et al. A review and evaluation of the state-of-the-art in PV solar power forecasting: techniques and optimization. Renew Sustain Energy Rev 2020;124:109792.
[42] Alamaniotis M, Ikonomopoulos A, Tsoukalas LH. A Pareto optimization approach of a Gaussian process ensemble for short-term load forecasting. In: 2011 16th International conference on intelligent system applications to power systems. IEEE; 2011.
[43] Gao WF, Liu SY. Improved artificial bee colony algorithm for global optimization. Inform Process Lett 2011;111(17):871–82.
[44] Perles L, et al. Diversity and seasonal dynamics of ticks on ring-tailed coatis Nasua nasua (Carnivora: Procyonidae) in two urban areas from Midwestern Brazil. Animals (Basel) 2022;12(3):293.
[45] Yu XD, et al. Wind power interval forecasting based on confidence interval optimization. Energies 2018;11(12):3336.
[46] Markovics D, Mayer MJ. Comparison of machine learning methods for photovoltaic power forecasting based on numerical weather prediction. Renew Sustain Energy Rev 2022;161:112364.
[47] Dolara A, et al. Weather-based machine learning technique for day-ahead wind power forecasting. In: 2017 IEEE 6th international conference on renewable energy research and applications (ICRERA). IEEE; 2017.
[48] Neshat M, et al. Wind turbine power output prediction using a new hybrid neuro-evolutionary method. Energy 2021;229:120617.
[49] Hao Y, Tian CS. A novel two-stage forecasting model based on error factor and ensemble method for multi-step wind power forecasting. Appl Energy 2019;238:368–83.
[50] do Nascimento Camelo H, et al. Innovative hybrid models for forecasting time series applied in wind generation based on the combination of time series models with artificial neural networks. Energy 2018;151:347–57.
[51] Guo Z, Chang C, Wang R. A novel method to downscale daily wind statistics to hourly wind data for wind erosion modelling. In: Geo-informatics in resource management and sustainable ecosystem. Springer; 2015. p. 611–9.
[52] Sun W, Liu MH, Liang Y. Wind speed forecasting based on FEEMD and LSSVM optimized by the bat algorithm. Energies 2015;8(7):6585–607.
[53] Tian C, Niu T, Wei W. Developing a wind power forecasting system based on deep learning with attention mechanism. Energy 2022;257:124750.
[54] Netsanet S, et al. Short-term PV power forecasting using variational mode decomposition integrated with ant colony optimization and neural network. Energy Rep 2022:8.
[55] Li Y, et al. Wind power forecasting considering data privacy protection: a federated deep reinforcement learning approach. Appl Energy 2023;329:120291.
[56] Al-qaness MA, et al. Boosted ANFIS model using augmented marine predator algorithm with mutation operators for wind power forecasting. Appl Energy 2022;314:118851.